

installation. It can be accessed from <https://hub.docker.com/>.

The millions of images available on Docker Hub can be downloaded and integrated for real world applications. These images are deployed under different categories related to the operating system, system architecture and technology.

To use a CentOS image from Docker Hub and run it on Ubuntu, type:

```
$ docker run -it centos /bin/bash
```

Docker on cloud

Docker is not only available for dedicated infrastructure but can also be deployed on cloud environments. A number of cloud platforms provide Docker hosting for real world and high performance applications.

There are many platforms on which Docker containers can be deployed.

A few of these are:

- Amazon Cloud
- Google Cloud
- Clever Cloud
- Microsoft Azure
- Kamatera
- HostPresto
- Red Hat OpenShift
- Vultr

Scope for research and development

Even though Docker is very popular for assorted use cases in enterprise level applications, there is huge scope for research in this domain. Researchers, practitioners and academicians can work on a number of problems related to new generation virtualisation, including:

- Migration and security aspects of Docker containers
- Privacy and integrity of Docker containers on cloud
- Virtual machine (VM) migration and placement algorithms
- Resource optimisation in new generation virtualisation technology
- Microservices management in virtualisation and cloud environment



Figure 4: Popular Docker images on public repository of Docker Hub

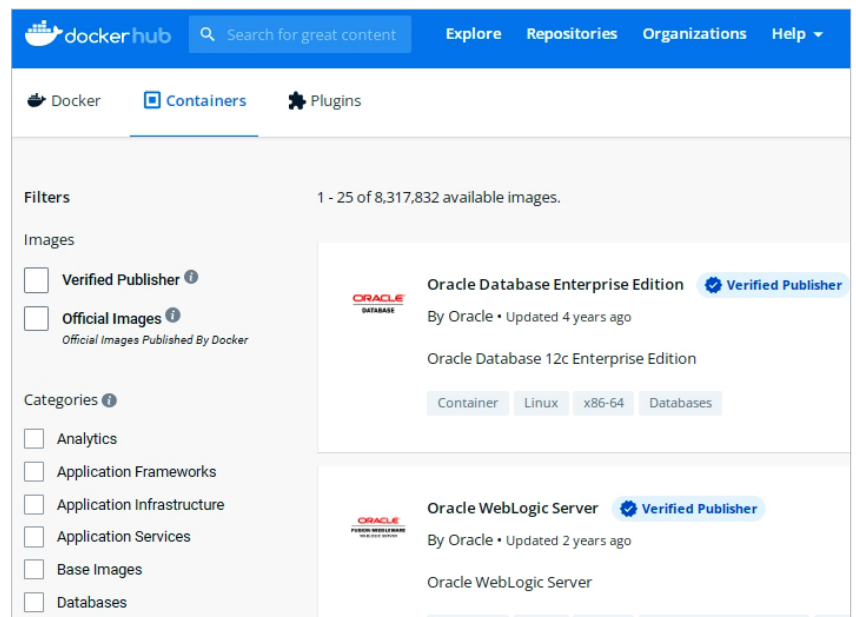


Figure 5: Searching Docker images on Docker Hub

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